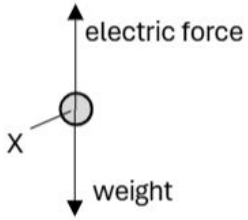
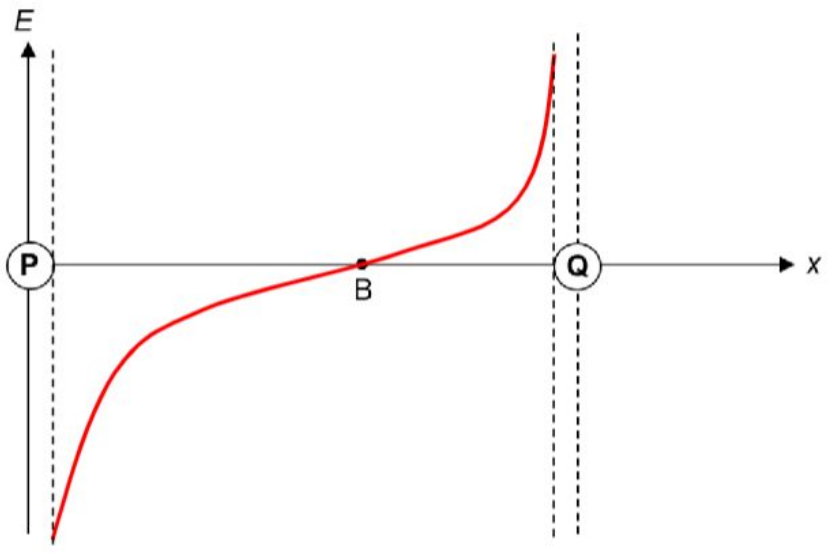
7 (a)(i)	 1 mark for both forces correctly labelled with correct directions	 **A1**
(ii)	Electric field strength is defined as the **force per unit positive charge** **acting on a small stationary charge placed** at a point **in the electric field**.	**B1** **B1**
(iii)	$F_E = F_G$ $qE = mg$ $E = mg / q$ $= (1.00 \times 10^{-6})\ (9.81) / (1.60 \times 10^{-12})$ $= 6.1 \times 10^6$ N C^{-1} (2 sf)	 **C1** **M1**
(iv)	$E = \Delta V / d$ $\Delta V = E\ d$ $= (6.1 \times 10^6)\ (5 \times 10^{-3})$ $= 30.500$ kV $\Delta V = V_{bottom} - V_{top}$ $V_{bottom} = 30.500 + (-15.0)$ $=$ <u>15.5 kV</u> (2 sf)	 **C1** **A1**
(v)(A)	When separation between plates increases, the **electric field strength decreases**. This results in a decrease in the electric force (acting upwards) on the particle. The **weight of the particle** (acting downwards) is now **greater than the electric force**, which will cause the particle's **path to be deflected downwards / towards the bottom plate**.	**B1** **B1**
(v)(B)	There is **no change to the path** of the particle as the path is independent of the horizontal velocity of the particle.	**B1**

7(b)(i)	The **electric field strength is equal** to the **negative potential gradient**.	**B1**
(ii)	 1 mark for correct shape of the graph 1 mark for the cutting point B at $E = 0$	 **A1** **A1**
(iii)	Work done $= q\,\Delta V$ $= (1.2 \times 10^{-9})\,(V_C - V_A)$ $= (1.2 \times 10^{-9})\,[-4 - (-10)]$ $= \underline{7.2 \times 10^{-9}}$ J	 **C1** **A1**
(iv)	**Between A and B**, the potential gradient is positive, which means that the **resultant electric field strength is negative and pointing towards P**. **Between B and C**, the potential gradient is negative, which means that the **resultant electric field strength is positive and pointing towards Q**. The only way for the above to happen is that **both charges on P and Q are negative**. Alternative answer: can accept other reasonable answers that make use of data from Fig. 7.2 or 7.3.	**M1** **A1**
(v)	From Fig. 7.2, $V_B = -1.90$ V. $V_B = V_{\text{due to P}} + V_{\text{due to Q}}$ $-1.90 = \frac{1.5Q}{4\pi\varepsilon_0(0.0275)} + \frac{Q}{4\pi\varepsilon_0(0.0225)}$ Solving for Q, Charge on sphere Q = $\underline{2.13 \times 10^{-12}}$ C Charge on sphere P = $\underline{3.20 \times 10^{-12}}$ C	 **C1** **C1** **A1**